

Name/UID _____

An Aggie does not lie, cheat or steal or tolerate those who do!

Please answer the 6 multiple-choice questions and the written questions that follow. Good luck! This is a closed-book exam. You may use one two-sided, letter-sized sheet of notes and a scientific calculator. Smartphones, tablets, smartwatches, laptops, and notebooks are not allowed.

1. (0.33pt) What does the posterior distribution represent?
 - (a) The joint probability of all hypotheses.
 - (b) The prior distribution normalized.
 - (c) The probability distribution over world states given the data.
 - (d) The probability of observations given a hypothesis.
2. (0.33pt) What is the correct definition of likelihood?
 - (a) The probability of an observation under a hypothesized state of the world.
 - (b) The ratio of posterior odds.
 - (c) The expected value of a utility function.
 - (d) The probability of the hypothesis given the observation.
3. (0.33pt) In the speech recognition example (“fork handles” vs “four candles”), why does Bayesian inference favor one interpretation over the other?
 - (a) Because the acoustic likelihoods are very different.
 - (b) Because the priors make one interpretation far more likely.
 - (c) Because the normalization constant cancels out one hypothesis.
 - (d) Because noise eliminates one option.
4. (0.33pt) Which of the following best describes Bayes’ theorem?
 - (a) A method for maximizing likelihood functions.
 - (b) A rule for updating probabilities of hypotheses given new data.
 - (c) A normalization step for probability distributions.
 - (d) A principle for rejecting incorrect hypotheses.
5. (0.33pt) In Bayesian inference, what is the main difference between likelihood and posterior?
 - (a) The likelihood measures data fit under a hypothesis, while the posterior updates beliefs about hypotheses.
 - (b) The likelihood includes priors, but the posterior does not.
 - (c) The likelihood is always larger than the posterior.
 - (d) The posterior is used only when priors are uniform.
6. (0.33pt) Why is Bayesian inference often more robust in uncertain environments than maximum likelihood estimation (MLE)?
 - (a) MLE always ignores the data.
 - (b) MLE double-counts observations.
 - (c) Bayesian inference avoids probability distributions entirely.
 - (d) Bayesian inference incorporates prior knowledge and uncertainty.

7. Let's suppose that you are an air travel along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage. A recording piped through the speakers reminds you that *"Many bags look alike. Please check your bag carefully before exiting the terminal."* Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case used by 5 percent of all travelers. Given your distance from the luggage chute, you can discern only the shape, size and color of the bags that emerge, not any detailed markings on the bags.



- a. (1pt) Let's suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain.

Answer:

$$H_1 : \text{the bag is mine,} \quad P(H_1) = 0.01,$$

$$H_2 : \text{the bag is not mine,} \quad P(H_2) = 0.99.$$

$$P(\text{obs} \mid H_1) = 1, \quad P(\text{obs} \mid H_2) = 0.05.$$

$$\begin{aligned} P(H_1 \mid \text{obs}) &= \frac{P(\text{obs} \mid H_1)P(H_1)}{P(\text{obs} \mid H_1)P(H_1) + P(\text{obs} \mid H_2)P(H_2)} \\ &= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168 \end{aligned}$$

16.8% chance of being my bag if the first bag to come looks like mine.

- b. (1pt) What is the probability that your bag is next when only 2 bags are left to come off the airplane?

Answer:

$$\begin{aligned} P(H_1 \mid \text{obs}) &= \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining} - 1}{\text{Total of bags remaining}}} \\ &\quad (\text{if 10 bags remain}) \\ &= \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.05 \cdot \frac{1}{2}} \\ &= \frac{1}{1 + 0.05 \cdot 1} = \frac{1}{1} \approx 0.952 \end{aligned}$$

4.8% chance that the last bag will not be mine if the one before the last one looks like mine.

8. Suppose you are waiting to catch a particular bus in a city that has just ten bus routes; the route followed by each bus is indicated by an integer in the corner of its front display. You see the bus below from a distance, and naturally wonder whether this is the bus you are waiting for.



- a. (1pt) Based on the visual image of the difficult-to-discern bus route number, and your intuitive understanding of how different route numbers might appear, construct a plausible likelihood function that plots $p(\text{visual observation} \mid \text{hypothesized bus route})$, for all numbers from 1 to 10. (Assume that the written destination does not mean anything to you.)

Answer: Likelihood function: from the blurry image of the bus, the plausible numbers are 3, 4, 5, 6. We assign likelihoods as follows:

$$p(\text{obs} \mid k) = \begin{cases} 0.35 & k = 3, \\ 0.35 & k = 4, \\ 0.15 & k = 5, \\ 0.10 & k = 6, \\ 0.05 & \text{otherwise.} \end{cases}$$

- b. (1pt) As it turns out, you happen to know that only buses 3, 4, 5, and 6 travel down the street you are on. Furthermore, you know that buses 3 and 4 come twice as frequently as buses 5 and 6. Based on this background knowledge, construct your prior distribution for the bus number.

Answer: Prior distribution: we are told that only buses 3, 4, 5, 6 travel on this street, and that buses 3, 4 arrive twice as often as buses 5, 6. Assigning relative weights:

$$w(3) = 2, \quad w(4) = 2, \quad w(5) = 1, \quad w(6) = 1,$$

with total weight 6, the prior probabilities are

$$p(3) = \frac{2}{6} = \frac{1}{3}, \quad p(4) = \frac{1}{3}, \quad p(5) = \frac{1}{6}, \quad p(6) = \frac{1}{6}.$$

- c. (1pt) Use Bayes' rule to calculate your posterior distribution for the number of the bus.

Answer: Posterior distribution (Bayes' rule):

$$p(k \mid \text{obs}) = \frac{p(\text{obs} \mid k) p(k)}{\sum_j p(\text{obs} \mid j) p(j)}.$$

Compute the unnormalized values:

$$3 : 0.35 \times \frac{1}{3} = 0.1167, \quad 4 : 0.35 \times \frac{1}{3} = 0.1167, \quad 5 : 0.15 \times \frac{1}{6} = 0.0250, \quad 6 : 0.10 \times \frac{1}{6} = 0.0167.$$

The denominator is

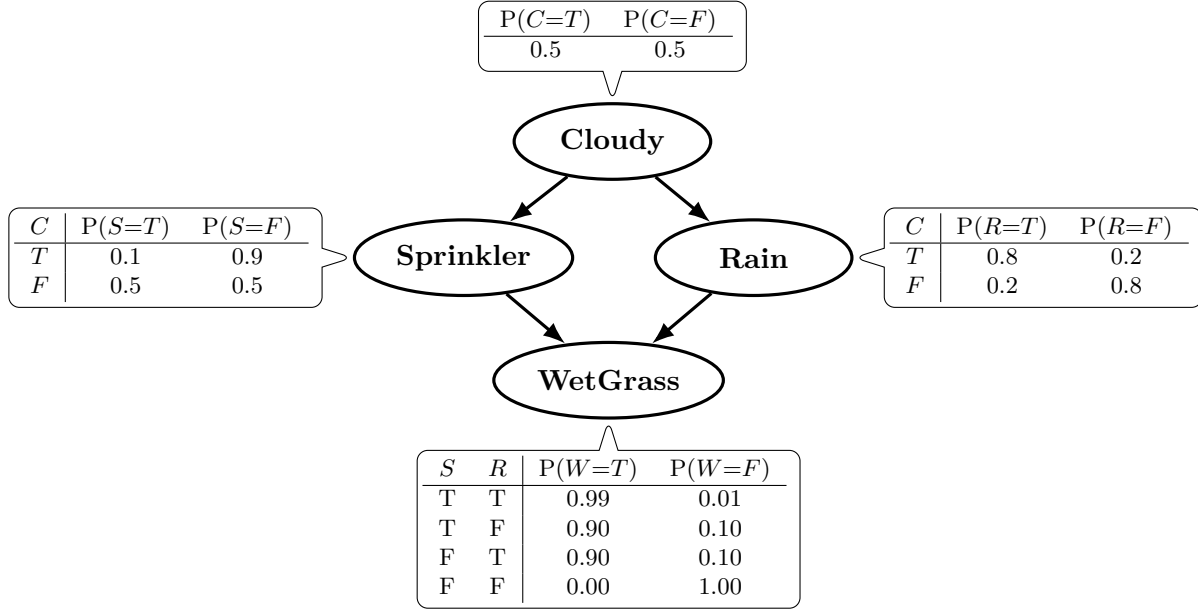
$$Z = 0.1167 + 0.1167 + 0.0250 + 0.0167 = 0.275.$$

Normalizing, we obtain the posterior:

$$\begin{aligned} p(3 \mid \text{obs}) &= \frac{0.1167}{0.275} \approx 0.424, & p(4 \mid \text{obs}) &= \frac{0.1167}{0.275} \approx 0.424, \\ p(5 \mid \text{obs}) &= \frac{0.0250}{0.275} \approx 0.091, & p(6 \mid \text{obs}) &= \frac{0.0167}{0.275} \approx 0.061. \end{aligned}$$

Thus the bus is almost equally likely to be number 3 or 4, smaller chances of being 5 or 6.

9. Using the graph below, please answer the questions and explain how to apply Bayesian inference; identify the conditional probabilities from the graph, and show each step, including total probability and any conditional dependencies.



C = Cloudy, S = Sprinkler, R = Rain, W = WetGrass, T = True, F = False.

- a. (1pt) Calculate the marginal probability that the grass is wet.

Answer: Factorization (from the DAG and conditional independence $S \perp R \mid C$):

$$P(C, S, R, W) = P(C) P(S \mid C) P(R \mid C) P(W \mid S, R)$$

CPTs:

$$\begin{aligned}
 P(C=T) &= P(C=F) = 0.5, \\
 P(R=T \mid C=T) &= 0.8, \quad P(R=T \mid C=F) = 0.2, \\
 P(S=T \mid C=T) &= 0.1, \quad P(S=T \mid C=F) = 0.5, \\
 P(W=T \mid S=T, R=T) &= 0.99, \quad P(W=T \mid S=T, R=F) = 0.9, \\
 P(W=T \mid S=F, R=T) &= 0.9, \quad P(W=T \mid S=F, R=F) = 0.
 \end{aligned}$$

$$P(W=T) = \sum_{c \in \{T, F\}} \sum_{s \in \{T, F\}} \sum_{r \in \{T, F\}} P(C=c) P(S=s \mid C=c) P(R=r \mid C=c) P(W=T \mid S=s, R=r)$$

$$\begin{aligned}
 P(W=T) &= \underbrace{0.5 \cdot 0.1 \cdot 0.8 \cdot 0.99}_{C=T, S=T, R=T} + \underbrace{0.5 \cdot 0.1 \cdot 0.2 \cdot 0.90}_{C=T, S=T, R=F} \\
 &\quad + \underbrace{0.5 \cdot 0.9 \cdot 0.8 \cdot 0.90}_{C=T, S=F, R=T} + \underbrace{0.5 \cdot 0.9 \cdot 0.2 \cdot 0.00}_{C=T, S=F, R=F} \\
 &\quad + \underbrace{0.5 \cdot 0.5 \cdot 0.2 \cdot 0.99}_{C=F, S=T, R=T} + \underbrace{0.5 \cdot 0.5 \cdot 0.8 \cdot 0.90}_{C=F, S=T, R=F} \\
 &\quad + \underbrace{0.5 \cdot 0.5 \cdot 0.2 \cdot 0.90}_{C=F, S=F, R=T} + \underbrace{0.5 \cdot 0.5 \cdot 0.8 \cdot 0.00}_{C=F, S=F, R=F}
 \end{aligned}$$

$$\begin{aligned}
 &= 0.0396 + 0.0090 + 0.3240 + 0.0000 \\
 &\quad + 0.0495 + 0.1800 + 0.0450 + 0.0000 = 0.6471
 \end{aligned}$$

$$P(W=T) \approx 0.6471$$

- b. (1pt) Compute $P(\text{Cloudy} = \text{True} \mid \text{Sprinkler} = \text{True})$

Answer:

$$P(C = T \mid S = T) = \frac{P(S = T \mid C = T) P(C = T)}{P(S = T)}$$

$$\text{Denominator: } P(S = T) = P(S = T \mid C = T) P(C = T) + P(S = T \mid C = F) P(C = F)$$

$$P(S = T) = (0.1)(0.5) + (0.5)(0.5) = 0.05 + 0.25 = 0.30$$

$$\text{Numerator: } P(S = T \mid C = T) P(C = T) = (0.1)(0.5) = 0.05$$

$$P(C = T \mid S = T) = \frac{0.05}{0.30} = \frac{1}{6} \approx 0.167$$

$$\boxed{P(C = T \mid S = T) \approx 0.167}$$

- c. (1pt) Given that the grass is wet, compute the posterior probability that it is cloudy.

Answer:

$$P(C = T \mid W = T) = \frac{P(W = T \mid C = T) P(C = T)}{P(W = T)}$$

$$P(W = T \mid C = T) = \sum_{s \in \{T, F\}} \sum_{r \in \{T, F\}} P(S = s \mid C = T) P(R = r \mid C = T) P(W = T \mid S = s, R = r)$$

$$\begin{aligned} P(W = T \mid C = T) &= (0.1)(0.8)(0.99) + (0.1)(0.2)(0.90) \\ &\quad + (0.9)(0.8)(0.90) + (0.9)(0.2)(0.00) \\ &= 0.0792 + 0.0180 + 0.6480 + 0.0000 = 0.7452 \end{aligned}$$

$$P(W = T) = 0.6471 \quad (\text{computed earlier})$$

$$P(C = T \mid W = T) = \frac{(0.7452)(0.5)}{0.6471} = \frac{0.3726}{0.6471} \approx 0.576$$

$$\boxed{P(C = T \mid W = T) \approx 0.576}$$

$$P(C = F \mid W = T) = \frac{P(W = T \mid C = F) P(C = F)}{P(W = T)}$$

$$P(W = T \mid C = F) = \sum_{s \in \{T, F\}} \sum_{r \in \{T, F\}} P(S = s \mid C = F) P(R = r \mid C = F) P(W = T \mid S = s, R = r)$$

$$\begin{aligned} P(W = T \mid C = F) &= (0.5)(0.2)(0.99) + (0.5)(0.8)(0.90) \\ &\quad + (0.5)(0.2)(0.90) + (0.5)(0.8)(0.00) \\ &= 0.0990 + 0.3600 + 0.0900 + 0.0000 = 0.5490 \end{aligned}$$

$$P(W = T) = 0.6471 \quad (\text{computed earlier})$$

$$\begin{aligned} P(C = F \mid W = T) &= \frac{(0.5490)(0.5)}{0.6471} \\ &= \frac{0.2745}{0.6471} \\ &\approx 0.424 \end{aligned}$$

$$\boxed{P(C = F \mid W = T) \approx 0.424}$$

$$\text{Check: } P(C = T \mid W = T) + P(C = F \mid W = T) = 0.576 + 0.424 = 1.000$$